

Subset-sum landscapes under deformed and random measures: the biased invariant $\Gamma^{(q)}$ and unconditional rigidity

(working title — scaffolding, round 100)

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August 9, 2026

Abstract

[**TODO** written last.] Papers 1–3 fixed both the sequence and the measure. This paper moves the measure along two axes — bias the coin, and randomise the sequence — and collects the two open questions recorded in paper 3.

1 Introduction

[**TODO** full introduction.] The two deformation axes, and why each is worth taking:

- *Bias.* The classification theorem of paper 1 is a statement about which elements lie on which side of a target; it says nothing about how configurations are weighted. So the whole combinatorial layer survives verbatim when the uniform measure is replaced by the Bernoulli(q) product measure, and the gap series deforms into an explicit one-parameter family. §2 does this, and the answer has a shape worth stating in the introduction: **the fair coin gives the least rugged landscape**.
- *Randomness.* For a random odd sequence the minor-arc input that cost paper 2 fifty rounds is replaced by a second-moment bound. §4 is therefore expected to carry *no ineffective constant at all* — the first such statement in the series. [**TODO** write; see §4.]

2 The biased invariant

Let P_q be the product measure under which each $a_i \in A$ is present independently with probability $q \in (0, 1)$, and write $\text{lm}_q(n)$, $\text{deg}_q(n)$ for the P_q -measures of the sets of strict local minima and of ground states at the target n . The target is taken at the q -tilted centre $n_q = \lfloor qT \rfloor$.

Definition 2.1 (biased gap series).

$$\Gamma^{(q)}(A) = 1 + \sum_{d \geq 1} \left[q^{N_d} + (1 - q)^{N_d} \right].$$

At $q = \frac{1}{2}$ every bracket is $2 \cdot 2^{-N_d}$ and $\Gamma^{(1/2)}$ is the window series W_D of paper 1, hence $\Gamma(A) + (2D + 1)2^{-k}$.

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Remark 2.2 (why the classification survives the deformation). Paper 1’s classification is measure-free: at error d , an overshooting configuration is a strict local minimum exactly when every $a \leq 2d$ is *excluded*, and an undershooting one exactly when every $a \leq 2d$ is *included*. Under P_q those two events have probability $(1-q)^{N_d}$ and q^{N_d} , which is Definition 2.1. Writing $r_B^q(m) = \sum_{U \subseteq B, \sigma(U)=m} q^{|U|}(1-q)^{|B|-|U|}$ and factorising P_q across I_d and B_d ,

$$\text{over}_q = (1-q)^{N_d} r_{B_d}^q(n+d), \quad \text{under}_q = q^{N_d} r_{B_d}^q(n-d-\sigma_d), \quad \text{deg}_q = \sum_{J \subseteq I_d} q^{|J|} (1-q)^{N_d-|J|} r_{B_d}^q(n-\sigma(J)),$$

and the binomial sum over J is exactly 1, so a flat r^q gives the d -th term of Definition 2.1. [\[derived; the assignment verified per stratum, see Remark 2.8\]](#)

2.1 The fair coin is the flattest

Lemma 2.3 (termwise minimality). *For every integer $N \geq 0$ the function $f_N(q) = q^N + (1-q)^N$ on $(0,1)$ is symmetric about $q = \frac{1}{2}$ and convex, and $f_N(q) \geq f_N(\frac{1}{2}) = 2^{1-N}$, with strict inequality for $q \neq \frac{1}{2}$ as soon as $N \geq 2$.*

Proof. $f_N(1-q) = f_N(q)$ is immediate. For $N \geq 2$, $f'_N(q) = N[q^{N-1} - (1-q)^{N-1}]$ and $f''_N(q) = N(N-1)[q^{N-2} + (1-q)^{N-2}] > 0$, so f_N is strictly convex; and $t \mapsto t^{N-1}$ is strictly increasing, so $f'_N < 0$ on $(0, \frac{1}{2})$ and $f'_N > 0$ on $(\frac{1}{2}, 1)$. For $N \in \{0, 1\}$, f_N is the constant 2 or 1. $\square$

Remark 2.4 (formally verified). Lemma 2.3 is machine-checked. `termwise_min` gives the inequality for every N and every $q \in [0, 1]$, `termwise_min_strict` its strictness for $N \geq 2$ and $q \neq \frac{1}{2}$, and `termwise_min_iff` the equality case as a biconditional; the normalisation $2 \cdot (\frac{1}{2})^N = 2^{1-N}$ is `two_mul_half_pow`. The non-strict bound is proved twice, once from convexity of x^N on $[0, \infty)$ and once from Bernoulli’s inequality (`termwise_min'`), so the two proofs check the *statement* against each other and not merely the tactic script. No sorry, and no axioms beyond `propext`, `Classical.choice`, `Quot.sound`. [\[Lean-verified; 20260809_TermwiseMin.lean\]](#)

Theorem 2.5 (the fair coin minimises the invariant). *For every finite increasing sequence A of odd positive integers with $|A| \geq 2$ and every $q \in (0, 1)$,*

$$\Gamma^{(q)}(A) \geq \Gamma^{(1/2)}(A),$$

with equality if and only if $q = \frac{1}{2}$.

Proof. Sum Lemma 2.3 over d . For strictness it suffices that some $d \leq D$ has $N_d \geq 2$: take any $d \geq (a_2 - 1)/2$, which is at most $D = (a_k - 1)/2$ because $a_2 \leq a_k$. $\square$

Remark 2.6 (what the theorem says). $q(1-q)$ is the per-element variance of the coin. Biasing the coin in either direction lowers that variance and raises the invariant: **the more predictable the coin, the more rugged the landscape**. The effect is exactly symmetric in $q \leftrightarrow 1-q$, as it must be, since complementing every configuration exchanges q with $1-q$ and n with $T-n$.

Corollary 2.7 (closed form for the odd numbers). *For $A = (1, 3, \dots, 2k-1)$ one has $N_d = d$ for $d \leq k$, so in the limit $k \rightarrow \infty$*

$$\Gamma^{(q)} = 1 + \frac{q}{1-q} + \frac{1-q}{q} = \frac{1}{q(1-q)} - 1 \geq 3,$$

with equality only at $q = \frac{1}{2}$: the invariant of the odd numbers is the reciprocal of the coin’s variance, minus one. [\[verified against the summed series to \$5.3 \cdot 10^{-15}\$ over \$q = 0.10, \dots, 0.80\$; e5_r098\]](#)

2.2 Verification

[experimentally confirmed] Measured lm_q/deg_q at n_q against the point prediction of Definition 2.1, relative error:

profile	q	$k = 60$	$k = 100$	$k = 140$
odds	0.20	$1.00 \cdot 10^{-2}$	$3.16 \cdot 10^{-3}$	$1.49 \cdot 10^{-3}$
odds	0.30	$2.63 \cdot 10^{-3}$	$8.02 \cdot 10^{-4}$	$3.75 \cdot 10^{-4}$
odds	0.50	$2.73 \cdot 10^{-4}$	$5.99 \cdot 10^{-5}$	$2.19 \cdot 10^{-5}$
primes	0.30	$2.18 \cdot 10^{-3}$	$5.02 \cdot 10^{-4}$	$2.20 \cdot 10^{-4}$
primes	0.50	$1.73 \cdot 10^{-4}$	$2.26 \cdot 10^{-5}$	$7.32 \cdot 10^{-6}$

There are no free parameters. Two features: the error decays faster than $1/k$ (fitted exponent 2.25 at $q = 0.2$ to 3.0 at $q = 0.5$), and it grows monotonically as q leaves $\frac{1}{2}$, by a factor ≈ 40 from $q = 0.5$ to $q = 0.2$ at fixed k . The second is what the biased flatness hypothesis will have to quantify. [TODO state the biased (H); design after §2.3.]

Remark 2.8 (the assignment, and a test that could not decide it). Definition 2.1 is symmetric in $q \leftrightarrow 1 - q$, and so is the exact complementation identity $\text{lm}_q(n) = \text{lm}_{1-q}(T - n)$ — at the q -tilted centre, $T - qT = (1 - q)T$. Neither can therefore distinguish the assignment of q^{N_d} and $(1 - q)^{N_d}$ to the two strata: measured at $k = 100$, the ratio at $q = 0.4$ and at $q = 0.6$ agrees to every printed digit. The discriminating test is per stratum, over q/deg_q against $(1 - q)^{N_d}$ and under q/deg_q against q^{N_d} ; measured, both ratios lie in $[0.9978, 1.0018]$ over $d \leq 6$, whereas a swap would disagree by a factor of 161 at $d = 6$. [experimentally confirmed; e5_r098]

2.3 The bias breaks the reflection symmetry

[experimentally confirmed, shape] Off the q -tilted centre the deviation $\text{dev}_q = (\text{lm}_q/\text{deg}_q)/\Gamma^{(q)} - 1$ behaves differently at $q = \frac{1}{2}$ and away from it, and the reason is structural. Paper 3's transfer function is even in λ because the two strata sit at $\pm\delta_d$ about a common centre — the analytic form of $\text{lm}(n) = \text{lm}(T - n)$. Under bias that symmetry maps q to $1 - q$ rather than to itself, so the odd orders need not cancel, and the first-order coefficient is

$$L^{(q)}(A) = \sum_{d \geq 1} \delta_d [q^{N_d} - (1 - q)^{N_d}],$$

which is antisymmetric in $q \leftrightarrow 1 - q$ and vanishes identically at $q = \frac{1}{2}$. Measured for the odd numbers at $k = 80$: $L^{(q)} = -28.634, -8.843, 0, +8.843, +28.634$ at $q = 0.3, 0.4, 0.5, 0.6, 0.7$, and for the primes $-83.717, -23.677, 0, +23.677, +83.717$.

The data agree. Over the same range of targets, the spread (max over min) of $|\text{dev}_q - \text{dev}_q(0)|$ divided by λ_q and by λ_q^2 :

profile	q	divided by λ_q	divided by λ_q^2
odds	0.30	1.329	9.644
odds	0.50	5.091	1.001
primes	0.30	1.284	8.858

So the response is linear in λ under bias and quadratic at $q = \frac{1}{2}$, where $(\text{dev} - \text{dev}_0)/\lambda^2$ is constant to four digits (19.8186 to 19.8352 across a fivefold range of the target offset). [experimentally confirmed; e6_r100] [TODO the primes at $q = \frac{1}{2}$ are not a clean test in this

range: $\text{dev} - \text{dev}_0$ crosses zero inside it, and dropping the single point where it is $2.7 \cdot 10^{-7}$ leaves the quadratic ratio in $[45.06, 48.86]$. Re-measure on a target range that does not straddle the crossing.]

3 The biased transfer function

[**TODO** fable-5: derive $\Phi^{(q)}$ by the partition-function method of paper 3’s rem:mushift, with the biased normaliser. The two strata carry unequal weights q^{N_d} and $(1-q)^{N_d}$ and a q -shifted μ_d , so the cosh becomes a general combination of two exponentials; §2.3 says the surviving odd part has coefficient $L^{(q)}$, which is the first thing the derivation must reproduce.]

4 Random sequences, unconditionally

Before the theorem, the floor it has to stand on. Let A be drawn from the Cramér model on the odd integers: each odd $m \in [3, N]$ is included independently with probability $2/\log m$, and N is fixed by $\mathbb{E}|A| = k$. Nothing is conditioned, so $|A|$ fluctuates from instance to instance. That is deliberate: $kR_A(k)$ is the k -invariant quantity of paper 3’s rem:onek, so an ensemble of genuinely different sizes tests the $1/k$ law for free. Twenty instances at each of two sizes, each through the whole paper-3 pipeline with its own N_d and σ_d :

$\mathbb{E} A $	range of k	(H) series	mean kR	sd	sd/mean
120	101–135	133.79–320.79	214.96	10.58	4.92%
180	146–191	132.94–471.77	206.32	6.94	3.37%

at $x = 0.06$; $c_A = kR/100$. Three things, and the third is the one to keep. [\[experimentally confirmed; e7_r102\]](#)

(i) *The hypothesis holds on the ensemble.* All forty instances have a bounded (H) series and a window that tracks the primes’. This is a precondition, not a result: a theorem about random sequences is unconditional only if the random sequences satisfy (H) themselves.

(ii) *The constant concentrates.* The spread across instances is under 5% and falls with k , while the mean of kR moves by 4% between the two sizes — the $1/k$ law of paper 3 surviving an ensemble it was not fitted on.

(iii) *The primes sit inside the ensemble and the odd numbers do not.* Against the $k \approx 180$ ensemble (2.0632 ± 0.0694 in c_A), the primes score $z = -0.39$ and the odd numbers $z = -2.70$, at every one of the four targets. In this statistic the primes are indistinguishable from a random odd sequence of the same density; the odd numbers are a different object.

There is one place where the random model is *not* like the primes, and it is the place that matters for the minor arcs. Write $h_d(q) = \max_{(r,q)=1} (\prod_{a \in B_d} |\cos(\pi r a/q)|)^{1/|B_d|}$. Since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a , the modulus-4 floor $h_d(4) = 1/\sqrt{2}$ is deterministic, not merely universal. Paper 2’s vanishing lemma kills the modulus- $2m$ peak (m odd) as soon as B_d contains an element $\equiv 3 \pmod{6}$. A set of primes contains exactly one such element, namely 3, and 3 leaves B_d at $d \geq 2$, so the primes keep $h_d(6) = \sqrt{3}/2$; a random odd sequence contains about $|B_d|/3$ of them and loses the peak at every d . Measured over $q \leq 120$ at $d = 1, 2, 5$: every random instance has $\max_{q \geq 3} h_d(q) = 1/\sqrt{2}$ attained at $q = 4$, and the primes have $\sqrt{3}/2$ at $q = 6$. [\[experimentally confirmed; e7_r102\]](#)

So the random theorem should come out with the geometric rate $1/\sqrt{2} = 0.7071$ per element rather than the $\sqrt{3}/2 = 0.8660$ of papers 1–2: *the primes are the harder case, and the modulus-6 ripple is a structural feature of the primes that the random model destroys.*

[**TODO** fable-5: for random odd A the minor-arc input is a second-moment bound $\mathbb{E}|S_A|^2 = b$ plus Borel–Cantelli on a θ -net — no Kumchev, no Siegel–Walfisz, no Siegel constant. If it lands it is the trilogy’s first zero-ineffective-constant rigidity theorem. The measurements above are the floor it must clear, and they sharpen the brief: the rate to aim for is $1/\sqrt{2}$, not $\sqrt{3}/2$.]

5 Two questions inherited from paper 3

(i) *The c_d fold-in is settled, and it goes paper 3’s way.* The share of the residual that c_d carries falls like $k^{-2\alpha}$ — measured exponents 1.98, 2.35, 2.99 against predicted 2.00, ≈ 2.4 , 3.00 on the odd numbers, the primes and $\alpha = \frac{3}{2}$, with the squares already below the numerical floor. So Φ as printed is right for every larger k and c_d belongs to E_k permanently; paper 3’s rem:cd now says so. [experimentally confirmed, four profiles, $k \leq 300$; e8_r103]

(ii) *$c_A = c'_A$ is still open, but sharper.* Carried to $k = 300$ the ratio reaches 1.0064 (odd numbers), 1.0001 (primes), 0.9984 ($\alpha = \frac{3}{2}$), and the gap closes like $k^{-2.4}$ — faster than the $1/k$ law both constants sit on. [experimentally confirmed; e8_r103] [**TODO** the mechanism is still missing, and a measured rate of convergence is evidence about a limit, not a proof of one. F45 stands: substituting s for λ makes the residual three times worse.]

6 Outlook

[**TODO** the algorithmic reading — local search on structured instances — which is where this programme started. Outlook only; no claims.]

7 Honest scope

- *Lean-verified:* Lemma 2.3 in all three forms (Remark 2.4), including the equality case.
- *Proved:* Lemma 2.3, Theorem 2.5, Corollary 2.7; and Remark 2.2’s derivation of Definition 2.1 modulo the biased flatness hypothesis, which is not yet stated.
- *Experimentally confirmed, with the range stated:* the point prediction (§2.2, two profiles, five q , three k), the stratum assignment (Remark 2.8), the linear response under bias (§2.3).
- *Not yet written:* §§3–6, and the biased hypothesis that §2 needs to become a theorem.

References

- [1] K. Amauchi, *The gap series of an integer sequence*. Manuscript, 2026.
- [2] K. Amauchi, *Asymptotic flatness of subset-sum landscapes of primes*. Manuscript, 2026.
- [3] K. Amauchi, *The transfer function of subset-sum landscapes*. Manuscript, 2026.